Binary Parsers in Typescript

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Binary Parsers

1. Varian NMR data parser
2. Development Workflow
3. Related Projects

Binary Parsers?

What are binary parsers?

Binary Parsers?

What are binary parsers?

Why not Python?

Binary Parsers?

What are binary parsers?

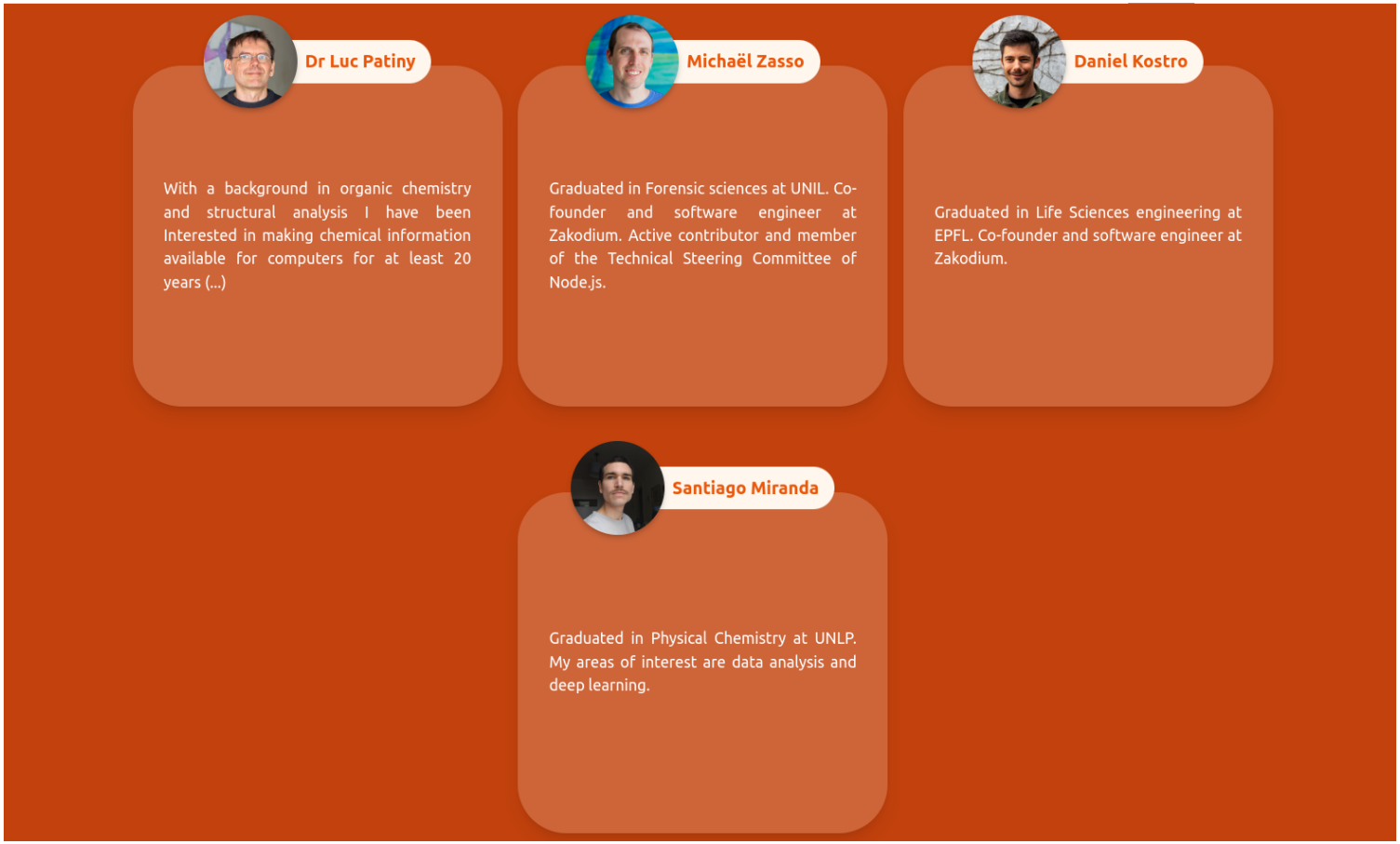
Why not Python?

Companies using these tools

EPFL cheminfo

.Z NMRium

Zakodium / NMRium

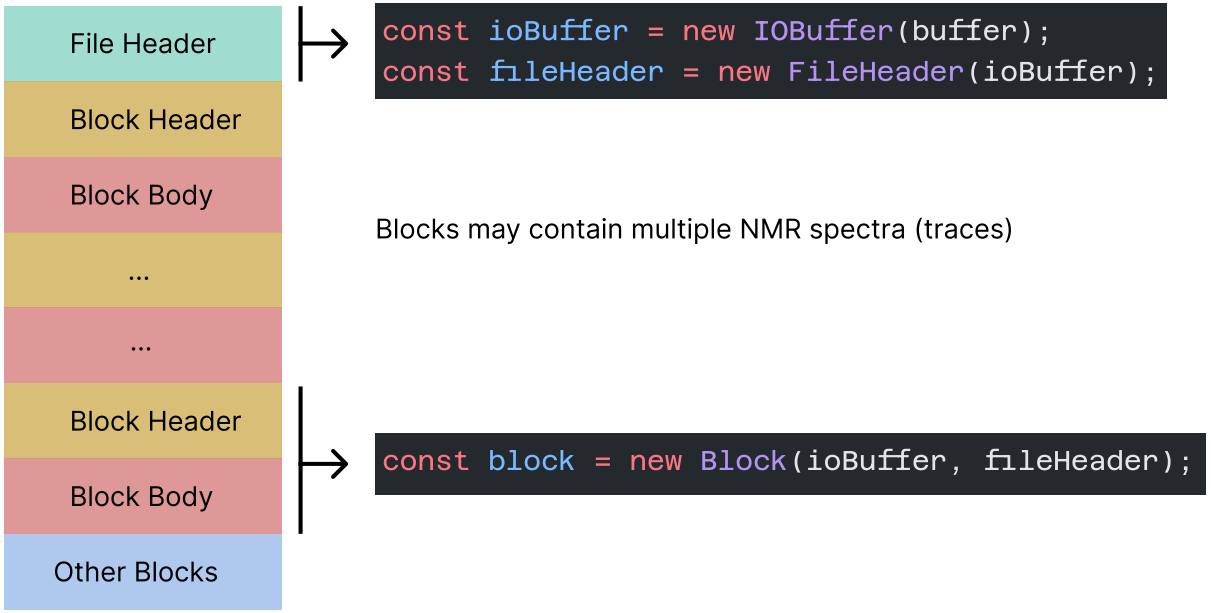


https://nmrium.org/about

Programming Languages

- **Typescript**
  - For Node and Browser
  - Different compilation targets
  - TF.js and ONNX
- **Python**
  - TF or PyTorch
  - Numpy and other libraries
- **Bash**
  - Automation and scripts

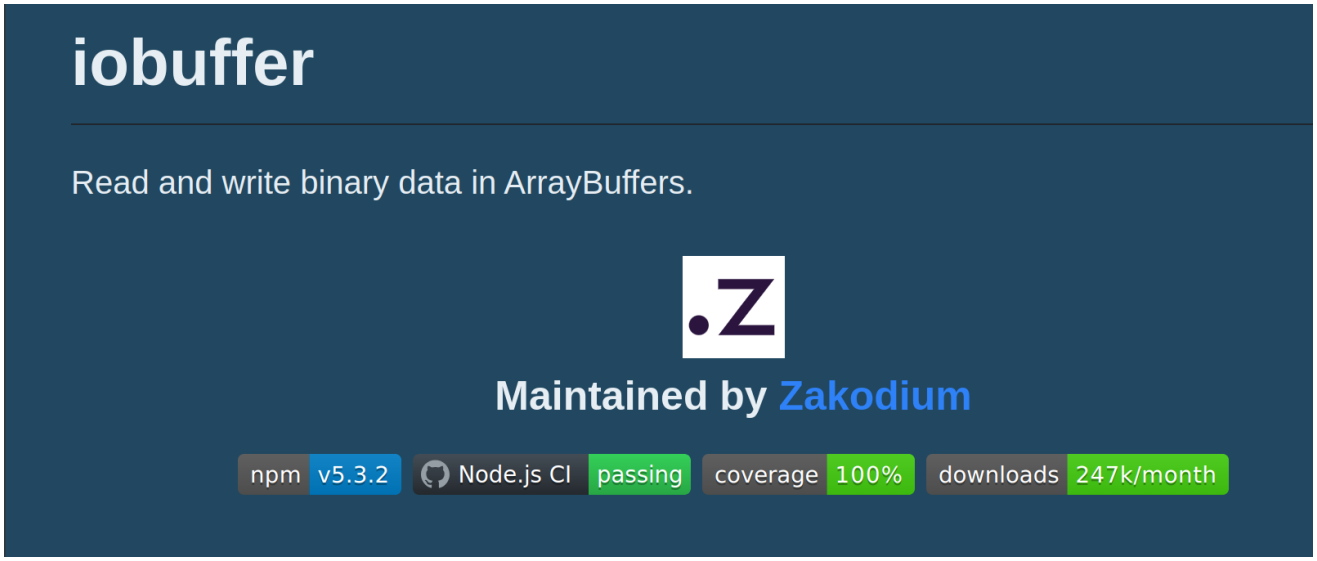
Varian Converter



Convert Varian NMR to JSON

IOBuffer

- IOBuffer: Read and write binary data in memory



https://github.com/image-js/iobuffer

IOBuffer

```
import { IOBuffer } from 'iobuffer';

const io = new IOBuffer() // offset 0
io.writeChars('Hello world') // offset now 11
  .writeUint32(42)
  .setBigEndian()
  .writeUint32(24)
  .skip(2)
  .rewind() // back to 0
  .readArray(11, "uint8") // read characters at once
```

Contributions: text decoder, readArrays of typed-data, host-endianness detection.

Development Workflow

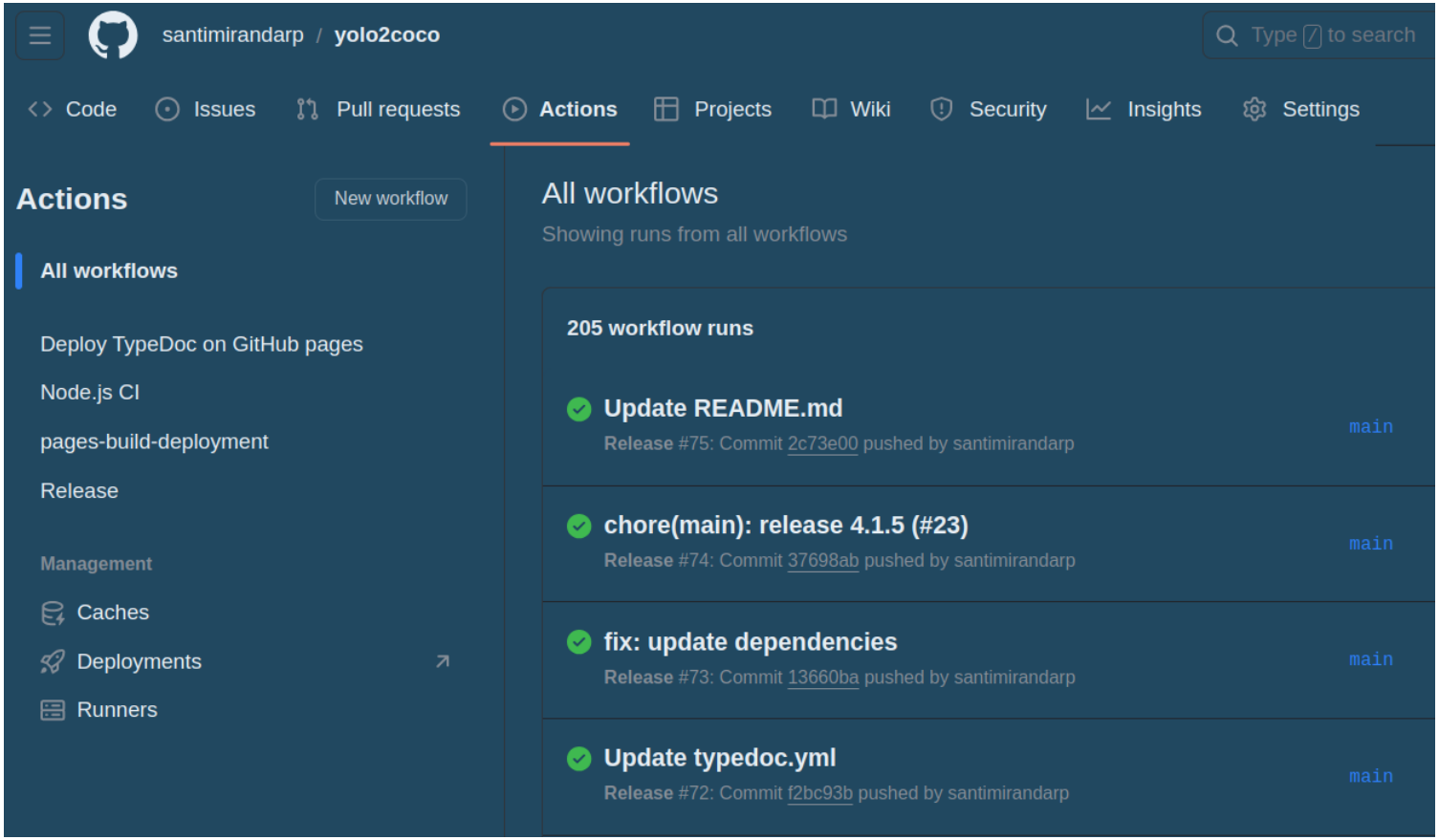
Development Workflow

Github & VSCode

VS Code and Workflows (CI/CD)

Tool	**Description**
Semantic Release	Changelog, Publish
Vitest	Testing
ESLint, Prettier	Lint and Format
Typescript	Type-check
Typedoc	Docs

Workflows


santimirandarp / yolo2coco
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Actions
New workflow
All workflows
Deploy TypeDoc on GitHub pages
Node.js CI
pages-build-deployment
Release
Management
Caches
Deployments
Runners
All workflows
Showing runs from all workflows
205 workflow runs
Update README.md
Release #75: Commit 2c73e00 pushed by santimirandarp
main
chore(main): release 4.1.5 (#23)
Release #74: Commit 37698ab pushed by santimirandarp
main
fix: update dependencies
Release #73: Commit 13660ba pushed by santimirandarp
main
Update typedoc.yml
Release #72: Commit f2bc93b pushed by santimirandarp
main

Development Workflow

VSCode

- Remote SSH
- NodeJS Debugger
  - 😞 Configure `launch.json`
  - 😀 Track all variables and find bugs

Development Workflow

Other Projects

- Pseudonymisation of Identity Documents

Other Projects

- mrz.zakodium.com

Other Projects

- mrz.zakodium.com

Other Projects

- **Pseudonymisation of Identity Documents**
  Deep Learning project (private repository)
- **nnls** (Non-negative least squares)
  https://github.com/mljs/nnls
- **fcnnls** (Fast Combinatorial Non-negative least squares)
  https://github.com/mljs/fcnnls
- **polynomial-regression**
  https://github.com/mljs/polynomial-regression
- **spc-parser**
  https://github.com/cheminfo/spc-parser
- **wdf-parser**
  https://github.com/cheminfo/wdf-parser

Thanks!